CRITIQUING TASK- VERSUS GOAL-ORIENTED APPROACHES: A CASE FOR MAKAM RECOGNITION

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ABSTRACT

Computational Musicology and Music Information Retrieval (MIR) address the core musical question under study from a different perspective, often combining top-down vs. bottom-up approaches. However, the evaluation metrics for MIR tend to capture the model accuracy in terms of the goal. [1] For instance, mode recognition is implemented with a goal to evaluate and compare melodic analysis approaches, but it is worth investigating if at all it lends itself as one befitting proxy task. [2] This is particularly relevant in non-Eurogenetic music repertoires where the grammatical rules are rather prescriptive. We employ methodologies that combine domain knowledge and data-driven optimisations as a possible way for a comprehensive understanding of these relationships. This is tested on Makam, one of the understudied corpora in MIR. We evaluate an array of feature-engineering methods on the largest mode recognition dataset curated for Ottoman-Turkish makam music, composed of 1000 recordings in 50 makams. We also address (ethno)musicology-driven tasks with a view to gathering more profound insights into this music, such as tuning, intonation, and melodic similarity. We aim to propose avenues to extend the study to makam characterisation over the mere goal of recognizing the mode, to better understand the (dis)similarity space and other plausible musically interesting facets.

1. INTRODUCTION

The melodic framework in many music traditions is often governed by the system of modes. A mode can be viewed as falling somewhere between a scale and a tune in terms of its defining grammar, which includes the tonal material, tonal hierarchy, and characteristic melodic movements [1–3]. While the function and the understanding of these frameworks are distinct from a culture-specific perspective, in a broader sense, they may be considered as the modes of the studied music culture. Some music traditions that can be considered modal are Indian art (raga) music, the Turkish/Arabic makam/maqam traditions, and the Gregorian church modes. Concerning the relevance of musical mode in non-Eurogenetic music repertoires, there are two contrasting viewpoints on whether to rethink or reject modal structure. While the former advocates for adapting the concept of mode as an underlying framework to systematise musical patterns, the latter tends to nullify the syntactic jargon of using a foreign language grammar (e.g., mode) to interpret literature in another (e.g., makam).

Makam/maqam is the melodic framework that consists of a system of scales defined by successive intervals, habitual melodic phrases, modulation pathways, ornamentation techniques and aesthetic conventions. It is used in Turkish (and Arabic, including the Middle East and Western Indian Ocean) music, providing a complex set of rules for compositions and performance. A typical subset of the repertoire — Ottoman-Turkish makam music (OTMM) — is well-established as a classical music tradition. Historically, there are a few hundred makams, whereas in practice, most of the repertoire is composed in one of the top 20 makams [4, 5]. In OTMM, melodies typically revolve around an initial tone and a final tone [6], where the final tone is referred to as being synonymous with tonic. There is no definitive reference frequency to tune the tonic. Recognizing makam is in itself a much more difficult task due to various characteristics such as heterophony and high variability in interpretations by musicians [6]. Moreover, from the pedagogy and practice perspective, recognizing the underlying makam with a unary label may not be interesting enough. A knowledge seeker (e.g., from the perspective of an anthropologist or ethnomusicologist) would rather gain wisdom on the characterisation of a makam and its discriminatory aspects to differentiate it from seemingly similar-sounding neighbouring makams.

In the realm of music information retrieval (MIR), mode recognition as a task has been given considerable importance from the purview of this, lending itself as one befitting proxy to evaluate and compare melodic analy-

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[1] We define goal as optimization of evaluation metrics, e.g., accuracy.
[2] We define task as what a model is intended to learn, e.g., modal features such as intervals, note sequence, relative salience, etc.

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sis approaches. Such a list is lengthy, ranging from automatic transcription, tuning analysis, music discovery, music similarity and recommendation to computational (ethno)musicology applications [7, 8]. However, in most practical scenarios, mode recognition is not the central theme but just the first step evidencing the robustness of acoustic features and/or statistical models. In such a setup, the recognition accuracy tends to become just another optimisation function. This simply indicates that the ethos of mode recognition does not remain a task anymore but a 'goal'. In this work, we aim to critically address how such an approach can mislead the uninitiated audience by attributing the complexity of the musical characteristics to the shortcomings of computational models.

It is inevitable to have thorough interdisciplinary collaboration to address the specifics in their entirety. However, we will approach the first study in this direction from a corpus-based computational musicology paradigm. This means developing culture-aware music technologies combing data- and knowledge-driven methods. We aim to benchmark the reported state-of-the-art literature, improve baseline features and simulate intuitive models to contest them. Once the potential of our approach in achieving the 'goal' accuracy is established, we break away from supervised (classification) to unsupervised (clustering) learning to appreciate the nuances and facilitate a comprehensive understanding of the same realm, this time as a 'task'. The byproduct of such a work is to provide tools for several musicologically-relevant subtasks such as tuning and intonation analysis, temporal modelling, and so on. While musicological studies latch on qualitative and limited representative examples, empirical methods work at the level of larger corpora and are, therefore, particularly useful for information retrieval-based tasks where scalability and reproducibility are highly regarded, if not mandated. The extracted features facilitate the curation of large audio corpora not only by greatly reducing the time and effort spent on manual annotations but also by providing automatically extracted, reliable and reproducible information. It is to be noted that we are limiting the scope of this work to building on past work pertaining to aggregated feature representation disregarding any time information. Hence, sequence models or sophisticated deep learning models (e.g. [9]) are not included in the current discourse.

However, in such corpus-based studies that too are underrepresented in MIR, the features extracted from the data, [3] source code and the experimental results are not always shared, making it more inaccessible to reproduce or build on the literature. Thus the unavailability of public tools, datasets, and reproducible experimentation are major obstacles to computational music information research, especially if such relevant tasks have not been applied to studied music traditions earlier. The CompMusic project [4] contributed towards bridging this gap by creating open corpora and computational tools for several non-Eurogenetic music repertoires. This work builds on the dataset introduced in MORTY (MOde Recognition and Tonic Ydentification toolbox), which is the largest mode recognition dataset curated for OTMM, composed of 1000 recordings in 50 makams [10].

In this work, our contribution is two-fold. Firstly, we adapt a new feature called time-delayed melody surfaces (TDMS) from raga recognition to makam recognition that shows comparable results to that of the current state-of-the-art [11]. The second contribution is to establish a similarity space of makam melodic features that characterise the root cause of erroneous cases. The structure of the paper is as follows. Section 2 discusses relevant literature on mode recognition at large and a detailed review of makam recognition, more as a goal and not a task. Section 3 describes the methodological details on audio preprocessing, the dataset(s), and feature extraction/modeling. Next, the experimental details regarding the model architecture and evaluation strategies are discussed in Section 4.1. This essentially engulfs the 'goal'-oriented approaches and comparison with the state-of-the-art, followed by the discussion of an alternative paradigm of unsupervised learning. The latter aids in the 'task'-based approach and highlights the gained musicological insights from this study and possible avenues of extending to makam characterisation over merely recognizing the label. Finally, Section 5 summarises the contributions and poses the scope for further developments in the current study.

2. MODE RECOGNITION

There has been extensive interest in mode recognition in the last two decades; a good summary is presented in [12]. Most of this work focuses on culture-specific approaches for music traditions like OTMM [10, 11, 13, 14], Carnatic music [15–17], Hindustani music [18–20], Dastgah music [21–23] and medieval chants [24, 25]. A considerable portion of these studies is based on comparing pitch distributions [13, 15, 16, 18, 19, 26], which are shown to be reliable in the respective mode recognition task or, for that matter, goal per se. There also exist recent approaches that are based on characteristic melodic motif mining using network analysis [17, 20], aggregating note models using automatic transcription [27–30], or audio-score alignment [31, 32]. All of these methods have been designed specifically to address the studied music culture (with the exceptions of [20] and [10]), and they are not generalisable to other music cultures without considerable effort. Next, we present some of the specific literature on mode recognition that we base our analyses on.

Pitch Distributions (PD) and Pitch Class Distributions (PCD) have been the state-of-art feature for mode recognition tasks for a very long time [10, 12, 13, 19], irrespective of the fact that they completely disregard the temporal aspects of the melody, which are essential to a mode characterisation [33]. Karakurt et al. [10] applied a joint recognition of makam and tonic using PDs and PCDs. In the training phase, the authors used kNN classifiers with either single or multiple distributions per mode. Their best performing model achieved an accuracy of 71.8% on the OTMM

[3] Commercial audio recordings are generally difficult to be made public due to copyright issues.

[4] https://compmusic.upf.edu/

recognition dataset (explained in Section 3.1). Gulati et al. [20] proposed a novel feature for mode recognition in the context of Hindustani and Carnatic ragas, called the time-delayed melody surface (TDMS), which we will use in this study. Authors reported that TDMS-based models outperform PCD-based models of [10] in the raga recognition task.

Yeşiler et al. [14] used a Multilayer Perceptron (MLP) on pitch distribution of first and last sections together with overall distributions using a feature vector of 159 attributes (53-TET * 3 octaves). The highest accuracy reported is 75.6% on the OTMM recognition dataset with additional insights on relevant segments for better discriminability. Demirel et al. [11] advocate the advantage of using chroma features for the mode recognition task as it discards the need for automatic melody extraction of polyphonic audio, hence getting away with the imperfections thereof. Authors created makam templates from annotated data and used template matching using support vector machine (SVM) classifiers. The best performing model achieved an accuracy of 77% on the OTMM recognition dataset, which, to our knowledge, is the state-of-the-art in makam recognition applied to the OTMM corpus. Other works are not strictly mode recognition but use the framework of representation-cum-distance-measure for discriminating between allied raga-pairs [34, 35]. Section 4 discusses aspects we borrow from this methodology to quantify the classification errors from a clustering viewpoint.

3. METHODOLOGY

In [10], mode recognition is formally defined as classifying the mode of an audio fragment from a discrete set of modes. In the context of OTMM, the problem reduces to classifying the makam. Given that the mode recognition framework is already established and that we are benchmarking on literature applied to OTMM, we save some real estate assuming that the data (pre)processing and partial feature extractions will be exactly reproduced.

3.1 The OTMM Corpus and the Dataset

Considering the lack of open data sources for makam music, the CompMusic project gathered audio recordings, music scores and relevant metadata, and published in the public domain the *Dunya Ottoman-Turkish Makam Music Corpus* [5, 36], which is currently the most representative corpus for OTMM available for computational research purposes. From the corpus, [10] curated a test dataset of audio recordings with annotated makam and tonic, called the *Ottoman-Turkish makam music recognition dataset*. The dataset covers 20 commonly performed makams [5] composed of 1000 audio recordings. A single makam is performed in each recording (i.e. there are 50 recordings per makam). To the best of our knowledge, this dataset is the largest and the most comprehensive dataset for the evaluation of automatic makam recognition. Finally, the dataset has been used by other researchers [10, 11, 14] to demonstrate their methods, including the current state-of-the-art makam recognition approach [11].

We use the latest version of the dataset. [6] We use the pre-computed melody time-series (termed as "predominant melody" by [36]) provided in CompMusic Dunya. [7] The pitch is detected at a hop-size to sampling-rate ratio of 0.023 that translates to 23 ms intervals for 44.1 kHz sampled audio [37]. To compare across performances, it is crucial to normalise the melody with respect to the tonic frequency. For this study, we use manually curated tonic frequencies linked from the *Ottoman-Turkish tonic dataset*. [8] To avoid the effect of nonlinearity in the logarithmic Hz scale, we normalise the pitch time-series to a log-linear cents scale.

3.2 Feature Extraction and Modeling

The next step is to synthesise derived features from the raw predominant melody. These mid- or high-level features can be interpreted and mapped to musicological inferences. We follow an approach akin to Krumhansl's [38] to compute the histogram of pitch samples to construct the pitch-class distribution. The pitch values are octave-folded (0 — 1200 cents) and quantised into p bins of equal width. The bin centre is the arithmetic mean of the adjacent bin edges. The salience of each bin is proportional to the accumulated duration of the pitches within that bin. A probability distribution function is constructed where the area under the histogram sums to unity. Even though we use the equivalent of a PD method attributed to the high bin resolution, we converge to a PCD [18]. The PCD configuration is given in Section 4. The first row of Figure 1 shows PCDs computed from each Mahur, Rast, and Acemşiran in the dataset, with the average pitch at each bin drawn as a dashed line.

The next step is to construct a two-dimensional surface based on the concept of delay coordinates (also termed phase space embedding) [20]. The time-delayed melody surface (TDMS) is a compact representation that captures both the tonal and the temporal characteristics of melody, is robust to octave errors, also partially nullifies the relevance of melody transcription. We experiment with different parameters (See Section 4). The second row of Figure 1 shows TDMS averaged from the TDMS of all recordings in Mahur, Rast, and Acemşiran makams in the dataset. The horizontal and vertical trajectories indicate pitch transitions between the pitch classes. The isolated square shape-formation indicates a separation between the higher and lower tetrachords in the course of the melodic progression. In both PCD and TDMS features, makams Rast and Mahur are similar to a high degree. The PCD of makam Acemaşiran bears relatively small differences from the prior two; however, the TDMS representation manages to significantly differentiate itself via dif-

[5] Namely: Acemaşiran, Acemkürdi, Bestenigar, Beyati, Hicaz, Hicazkar, Hüseyni, Hüzzam, Karcığar, Kürdilihicazkar, Mahur, Muhayyer, Neva, Nihavent, Rast, Saba, Segah, Sultanıyegah, Suzinak, and Uşşak.

[6] **dlfm2016-fix1** — `https://zenodo.org/record/4883680`

[7] `https://dunya.compmusic.upf.edu`

[8] `https://zenodo.org/record/260038`

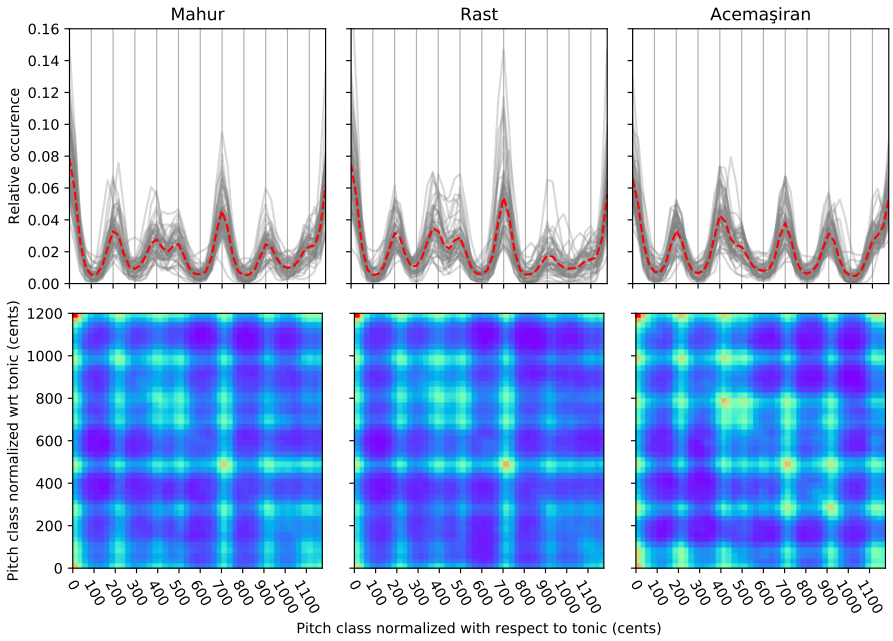


Figure 1. PCD (top row) and TDMS (bottom row) representations of Mahur (left column), Rast (middle column), and Acemaşiran (right column) makams.

ferences in melodic progression, captured through the delay coordinates. For example, in the TDMS representation, makams Mahur and Rast show clear transitions between 6th (900 cents), 7th scale degrees (1100 cents), and the tonic pitch class, whereas makam Acemşiran exhibits a progressions between 3rd (400 cents) and 6th scale degrees (900 cents).

4. EXPERIMENTS

Akin to the title of the paper, one of the main goals of the study is to treat makam recognition as a task and not a goal in itself. Our experiments are divided into two parts. The former addresses the 'goal'-oriented aspect, i.e., the best combination of features and classifiers in order to achieve the *optimal* accuracy. We also stress on intuitions why certain configuration of train-test partitioning would make more sense than others or why certain classifier is meant to 'learn' and not be totally data-proximity dependent. We report and discuss a subset of the results relevant to the optimal settings; the full experimental results are made available. [9]

For PCDs, we use the empirical "optimal" parameters for OTMM reported by [10], namely a bin resolution of 25 cents and Gaussian smoothing applied using a kernel width of 25 cents. We set the bin resolution of TDMS to 25 cents to compare with PCD, and grid-search time delay indices ($\in \{0.25, 0.5.1, 1.5, 2.5, 5\}$ seconds), compression exponents ($\in \{0.1, 0.25, 0.5, 0.75, 1\}$) and Gaussian smoothing kernel widths ($\in \{0, 12.5, 25, 50\}$ standard deviation in cents) in the classification experiments below to find the optimal configuration for OTMM.

[9] https://sertansenturk.com/work-research/ismir-2022-makam/

4.1 (Supervised) Classification

In line with the objective of supervised learning, i.e., to model the intra-class similarity and inter-class differences, the inherent 'goal' is to maximise classification accuracy. However, we carefully choose the feature set and classifier in order to suit the music theory. That is to say, we aim to incorporate knowledge constraints into mainstream data-driven computational models. We restrict ourselves to k-nearest neighbors [10, 12, 13, 19, 20], support vector machine [11, 12], multilayer perceptron [14] and logistic regression [12] that were extensively used in past mode recognition work.

Past studies used different cross-validation (CV) techniques such as leave-one-out CV [13], 10-fold CV [10, 19, 20] and nested k-fold CV [11, 14]. In our initial experiments, we compared nested 10-fold CV, 10-fold CV (without any unseen test set), and 10-times repeated shuffle split CV with 10% of the recordings reserved as test set for each repetition. We used stratified splits in all our experiments to keep the makam classes balanced and repeated each experiment 10 times. We report the mean & standard deviation of classification accuracy reported on the test set. We also compute a confusion matrix for each test set and aggregate it across all test sets in each repeated experiment per model. We report the results of the 10-times repeated shuffle split CV in the rest of the Section. Similar to [20], we observed that TDMS is robust in different time delay indices (between 0.5 and 2.5 seconds), kernel width (less than 25 standard deviations in cents) and compression exponent (above 0.25). For the rest of the experiments, we report results for TDMS with an "optimal" configuration of 1-second time-delay index, 12.5 cents of smoothing kernel width and a compression exponent of 0.5. Table 1

Model	PCD	TDMS
Support Vector Machine	$71.0 \mp 3.2\%$	$\mathbf{77.2 \mp 3.5}\%$
Multilayer Perceptron	$70.9 \mp 3.8\%$	$74.6 \mp 5.0\%$
k-nearest Neighbors	$68.2 \mp 3.4\%$	$70.2 \mp 3.1\%$
Logistic Regression	$66.8 \mp 3.9\%$	$75.5 \mp 4.1\%$

Table 1. Average $\mp$ standard deviation of classification accuracy for all feature-classifier combinations.

shows the mean and standard deviation in classification accuracy for PCD and TDMS using different models. In sum, TDMS with SVM works the best at par with the current state-of-the-art [11]. TDMS consistently performs better than PCD; statistical significance results are kept out of the scope of this work.

The associated confusion matrix for the optimal performing system is shown in Figure 2. Makams {Acemaşiran, Hicaz, Hüzzam} and {Bestenigar, Rast, Uşşak} are examples of highly discriminable and highly confused pairs respectively. The inferences from the confusion matrix are, however, limited to qualitative evaluation of the confused cases and a count of them. In the next Section, we propose a new approach to compute the pairwise distances in a clustering scenario which facilitates a quantitative evaluation of the proportion and magnitude of the confusions. This is, in a way, a manifestation of the recognition task wherein we model the inherent complexity in the data rather than the limitations of the method.

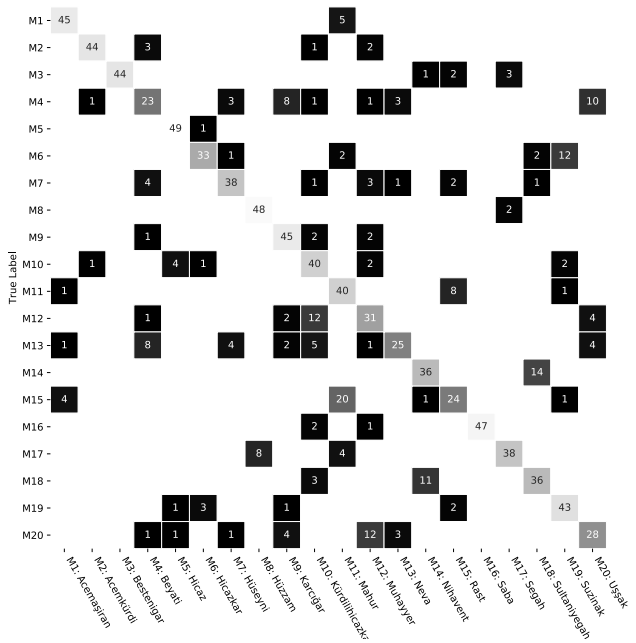


Figure 2. Aggregated confusions matrix for the optimal performing system: TDMS with SVM.

4.2 (Unsupervised) Clustering

To address the second half of the experiments, our focus moves to the 'task'-oriented results. This, we believe, is the highlight of the current contribution in terms of gaining/reconfirming musicological knowledge/concepts from the computational model that can feed back into the pedagogy and practices to further enrich the repertoire.

A typical mode recognition study would stop at this point after the goal accuracy is achieved [10, 11, 20]. Even though we report comparable results to that of the current state-of-the-art accuracy, we are keen on evaluating how much the representation-cum-distance-measure attribute to musicological insights. Over and beyond the error analysis, we stress validating the gap and reinforcing other possible avenues, so the complexity of the musical characteristics is not attributed to the shortcomings of computational models. One such way is to disregard the makam labels and study the melodic similarity space of relevant predictor features. Through these methods, we aim to verify whether the machine learning models indeed 'learn' what they are intended for. We present three complementary and supplementary retrieval scenarios to corroborate the 'task' details and bridge the gap that the best classification could achieve.

Hierarchical clustering: In the presence of theoretical grouping of makams, yet not having a prescription on the counts, it is practically impossible to set a k for a k-means clustering algorithm. However, hierarchical clustering seems to offer a dynamic solution in such scenarios. Here, each element is treated as distinct clusters at the lowest threshold, whereas there is a single giant cluster at the highest threshold. We present, in part of Figure 3, the dendrogram representation to capture the melodic similarity/grouping space across the 20 makams obtained from the hierarchical clustering of the TDMS features averaged from the 50 recordings per makam using Canberra distance. We use the Canberra distance to contrast with the hierarchical clustering reported in [14, see Figure 2] that was calculated from averaged pitch distributions, and keep the empirical experimentation of different distance metrics out of the scope. The groupings are shown in different colours, and the relative height where the tree elements merge is indicative of the normalised threshold. At the highest threshold, makam Saba isolates itself from the rest 19; this is indicative of the distinct nature of the pitch distribution captured through TDMS. Makam-pair (Rast, Mahur) show a very low distance, indicating high similarity in the feature space. A high distance between the (Hüzzam, Hicaz) pair, as shown in other figures, is also evident.

Cluster purity matrix: One supplementary way to capture all pairwise distances is through an unconventional method of computing a distance matrix with the salience function of cluster purity. This is broadly a homogeneity measure that evidences the quality of clustering. Any value close to 1 indicates perfect clustering, while 0.5 signifies random clustering. Out of the $\binom{20}{2} = 190$ distinct makam-pairs, 115 pairs show a cluster purity score ≥ 0.85, 73 other pairs bear cluster purity values in the range of (0.5,

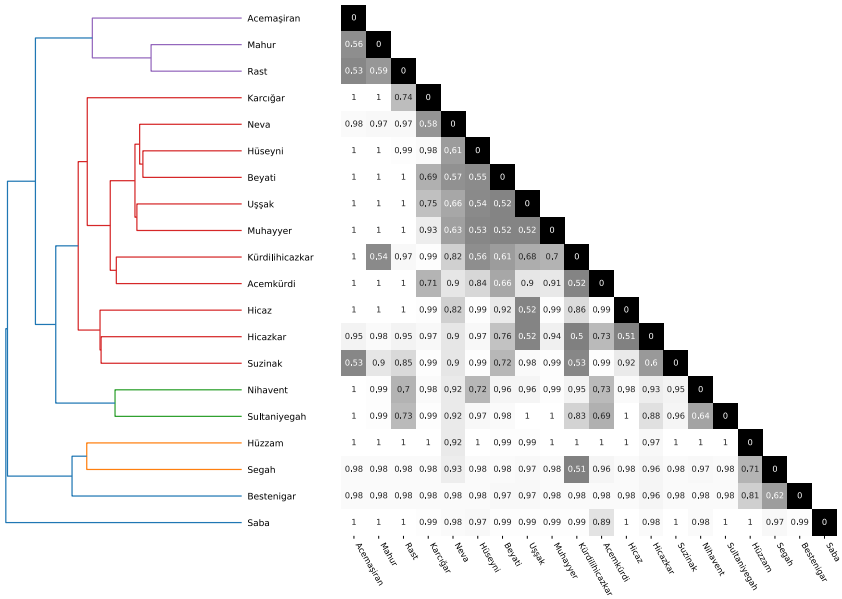


Figure 3. Left: dendrogram representation to capture the melodic similarity space across the 20 makams obtained from the hierarchical clustering of the TDMS. Right: the cluster purity matrix capturing the homogeneity of pairwise distances.

0.85), and none below a score of 0.5. We plot the pairwise purity scores partly in Figure 3, this representation has an intuitive inverse proportionality with the confusion matrix. The makam indices obtained from the dendrogram are aligned with that of the current matrix. It is intuitive to follow that the row corresponding to makam Saba (which is the most distinctive) has got the highest cluster purity (mode=1) with many other pairs, whereas the aforementioned confusable pairs yield a random clustering. This visualisation provides a complementary view of what is aggregated in the dendrogram.

Query retrieval score: The third scenario we present complementary to the clustering is the receiver-operated characteristics (ROC). We use the same two makam-pairs in the context of a query search for all possible matching and non-matching pairings out of each makams-pair. The ROCs in Figure 4 show the true positive rate versus the false positive rate achieved in the detection of non-matching makam pairs for the PCD (we consciously chose PCD over TDMS to introduce diversity) representations for four unique distance measures, inspired from [34]. The subplots correspond to makam-pairs (Hüzzam, Hicaz) and (Rast, Mahur), which are examples of highly discriminable and highly confused pairs, respectively. The ROC curves, the area under the curve (AUC) and equal error rate (EER) clearly indicate a better retrieval for the former, while the latter almost grazes the diagonal. We have inferences on why certain distance metric works better, but discussion on their relative performance is not directly related to the main narrative of this work and hence omitted [34, 39].

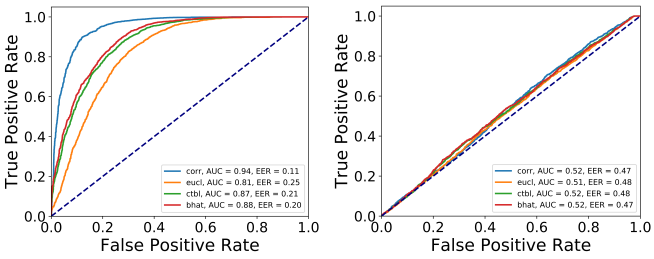


Figure 4. ROCs obtained using correlation (corr), euclidean (eucl), city-block (ctbl), and Bhattacharyya (bhat) distance from PCDs. Left: (Hüzzam, Hicaz) and right: (Rast, Mahur) makam-pairs.

5. CONCLUSION

We employed methodologies that combine domain knowledge and data-driven optimisations with a view to understanding the makam recognition 'task' in depth. We report comparable accuracy (77.2%) with the state-of-the-art [11] using the newly adapted TDMS feature with SVM. This achievement evidences the credential in our approach that provides us with a solid ground to argue the critique on goal- versus task-oriented approaches by comparing and contrasting. We have reported only the best-performing configuration in this paper [10] which will eventually be expanded to include temporal features and sequence models. In sum, we advocate that good supervised learning performance is a necessary but insufficient condition for a computational representation-cum-distance-measure to be considered informative for all purposes. As future work, we will incorporate convolutional networks and transformers to reproduce makam recognition on OTMM corpus to understand the potential of deep learning and the trade-off between goal and task involved. The application of such an approach aids in understanding music and other forms of sound culture and developing methodologies for cross-cultural mapping and comparing these materials.

[10] Array of alternate configurations: `https://sertansenturk.com/work-research/ismir-2022-makam/`

6. ACKNOWLEDGEMENTS

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7. REFERENCES

[1] H. S. Powers and R. Widdess, *India, subcontinent of*, 2nd ed., ser. New Grove Dictionary of Music. Macmillan, London, 2001, ch. III: Theory and practice of classical music, contributions to S. Sadie (ed.).

[2] H. S. Powers, F. Wiering, J. Porter, J. Cowdery, R. Widdess, R. Davis, and A. Marett, “Mode. grove music online,” 2008.

[3] K. K. Ganguli, “How do we ’See’ & ’Say’ a raga: A Perspective Canvas,” *Samakalika Sangeetham*, vol. 4, no. 2, pp. 112–119, Oct. 2013.

[4] T. Çevikoğli, “Klasik Türk müziğinin bugünkü sorunları,” in *Proceedings of International Congress of Asian and North African Studies (ICANAS 38’)*, Ankara, 2007.

[5] B. Uyar, H. S. Atli, S. Şentürk, B. Bozkurt, and X. Serra, “A corpus for computational research of Turkish makam music,” in *Proceedings of the 1st International Workshop on Digital Libraries for Musicology (DLfM 2014)*, London, UK, 2014, pp. 1–7.

[6] B. Bozkurt, R. Ayangil, and A. Holzapfel, “Computational analysis of Turkish makam music: Review of state-of-the-art and challenges,” *Journal of New Music Research*, vol. 43, no. 1, pp. 3–23, 2014.

[7] K. K. Ganguli and P. Rao, “On the perception of raga motifs by trained musicians,” *The Journal of the Acoustical Society of America*, vol. 145, no. 4, pp. 2418–2434, 2019.

[8] ——, “A study of variability in raga motifs in performance contexts,” *Journal of New Music Research*, vol. 50, no. 1, pp. 102–116, 2021.

[9] S. T. Madhusudhan and G. Chowdhary, “DEEPSRGM-sequence classification and ranking in Indian classical music with deep learning,” in *Proceedings of the 20th International Society for Music Information Retrieval Conference (ISMIR 2019)*, 2019, pp. 533–540.

[10] A. Karakurt, S. Şentürk, and X. Serra, “MORTY: A toolbox for mode recognition and tonic identification,” in *Proceedings of the 3rd International workshop on Digital Libraries for Musicology (DLfM 2016)*, New York, NY, 2016, pp. 9–16.

[11] E. Demirel, B. Bozkurt, and X. Serra, “Automatic makam recognition using chroma features,” in *Proceedings of the 8th International Workshop on Folk Music Analysis (FMA 2018)*. Thessaloniki, Greece: Aristotle University of Thessaloniki, 2018, pp. 19–24.

[12] G. K. Koduri, S. Gulati, P. Rao, and X. Serra, “Raga recognition based on pitch distribution methods,” *Journal of New Music Research*, vol. 41, no. 4, pp. 337–350, 2012.

[13] A. C. Gedik and B. Bozkurt, “Pitch-frequency histogram-based music information retrieval for Turkish music,” *Signal Processing*, vol. 90, no. 4, pp. 1049–1063, 2010.

[14] F. Yeşiler, B. Bozkurt, and X. Serra, “Makam recognition using extended pitch distribution features and multi-layer perceptrons,” in *Proceedings of the 15th Sound and Music Computing Conference (SMC 2018)*. Limassol, Cyprus: Cyprus University of Technology, 2018, pp. 249–253.

[15] P. Dighe, P. Agrawal, H. Karnick, S. Thota, and B. Raj, “Scale independent raga identification using chromagram patterns and swara based features,” in *Proceedings of IEEE International Conference on Multimedia and Expo Workshops (ICMEW)*, 2013, pp. 1–4.

[16] P. Dighe, H. Karnick, and B. Raj, “Swara histogram based structural analysis and identification of Indian classical ragas.” in *Proceedings of 14th International Society for Music Information Retrieval Conference (ISMIR 2013)*, 2013, pp. 35–40.

[17] S. Gulati, J. Serra, V. Ishwar, S. Şentürk, and X. Serra, “Phrase-based rāga recognition using vector space modeling,” in *Proceedings of IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP 2016)*. IEEE, 2016, pp. 66–70.

[18] P. Chordia and A. Rae, “Raag recognition using pitch-class and pitch-class dyad distributions.” in *Proceedings of 8th International Society for Music Information Retrieval Conference (ISMIR 2007)*, 2007, pp. 431–436.

[19] P. Chordia and S. Şentürk, “Joint recognition of raag and tonic in north Indian music,” *Computer Music Journal*, vol. 37, no. 3, pp. 82–98, 2013.

[20] S. Gulati, J. Serrà Julià, K. K. Ganguli, S. Sentürk, and X. Serra, “Time-delayed melody surfaces for rāga recognition,” in *Proceedings of the 17th International Society for Music Information Retrieval Conference (ISMIR 2016)*, New York, NY, 2016, pp. 751–757.

[21] S. Abdoli, “Iranian traditional music dastgah classification.” in *Proceedings of 12th International Society for Music Information Retrieval Conference (ISMIR 2011)*, 2011, pp. 275–280.

[22] P. Heydarian and D. Bainbridge, “Dastgàh recognition in Iranian music: Different features and optimized parameters,” in *Proceedings of 6th International Conference on Digital Libraries for Musicology (DLfM 2019)*, New York, NY, USA, 2019, p. 53–57.

[23] L. Cimarone, B. Bozkurt, and X. Serra, "Automatic dastgah recognition using Markov models," in *Proceedings of 14th International Symposium on Computer Music Multidisciplinary Research (CMMR 2019)*, Marseille, France, 10 2019, pp. 51–58.

[24] D. Huron and J. Veltman, "A cognitive approach to medieval mode: Evidence for an historical antecedent to the major/minor system," *Empirical Musicology Review*, vol. 1, no. 1, pp. 33–55, 2006.

[25] B. Cornelissen, W. Zuidema, and J. A. Burgoyne, "Mode classification and natural units in plainchant," in *Proceedings of the 21th International Society for Music Information Retrieval Conference (ISMIR 2020)*, 2020, pp. 869–875.

[26] K. K. Ganguli, S. Gulati, X. Serra, and P. Rao, "Data-driven exploration of melodic structures in Hindustani music," in *Proc. of the International Society for Music Information Retrieval (ISMIR)*, Aug. 2016, pp. 605–611, New York, USA.

[27] G. K. Koduri, V. Ishwar, J. Serrà, and X. Serra, "Intonation analysis of rāgas in Carnatic music," *Journal of New Music Research*, vol. 43, no. 1, pp. 72–93, 2014.

[28] K. K. Ganguli, A. Rastogi, V. Pandit, P. Kantan, and P. Rao, "Efficient melodic query based audio search for Hindustani vocal compositions," in *Proc. of Int. Soc. for Music Information Retrieval (ISMIR)*, Oct. 2015.

[29] K. K. Ganguli, A. Lele, S. Pinjani, P. Rao, A. Srinivasamurthy, and S. Gulati, "Melodic shape stylization for robust and efficient motif detection in hindustani vocal music," in *National Conference on Communications (NCC)*, 2017.

[30] K. K. Ganguli and P. Rao, "Discrimination of melodic patterns in Indian classical music," in *Proc. of National Conference on Communications (NCC)*, Feb. 2015.

[31] S. Şentürk, G. K. Koduri, and X. Serra, "A score-informed computational description of svaras using a statistical model," in *Proceedings of 13th Sound and Music Computing Conference (SMC 2016)*, Hamburg, Germany, 2016, pp. 427–433.

[32] J. C. Ross, A. Mishra, K. K. Ganguli, P. Bhattacharyya, and P. Rao, "Identifying raga similarity through embeddings learned from compositions' notation." in *Proc. of the International Society for Music Information Retrieval (ISMIR)*, 2017.

[33] S. Gulati, J. Serra, K. K. Ganguli, and X. Serra, "Landmark detection in Hindustani music melodies," in *Proc. of Int. Computer Music, Sound and Music Computing*, 2014, pp. 1062–1068, Athens, Greece.

[34] K. K. Ganguli and P. Rao, "On the distributional representation of ragas: experiments with allied raga pairs," *Transactions of the International Society for Music Information Retrieval*, vol. 1, no. 1, 2018.

[35] ——, "Towards computational modeling of the ungrammatical in a raga performance." in *Proceedings of the 18th International Society for Music Information Retrieval Conference (ISMIR 2017)*, Suzhou, China, 2017, pp. 39–45.

[36] S. Şentürk, "Computational analysis of audio recordings and music scores for the description and discovery of Ottoman-Turkish makam music," Ph.D. dissertation, Universitat Pompeu Fabra, Barcelona, Spain, December 2016.

[37] H. S. Atlı, B. Uyar, S. Şentürk, B. Bozkurt, and X. Serra, "Audio feature extraction for exploring Turkish makam music," in *Proceedings of 3rd International Conference on Audio Technologies for Music and Media (ATMM 2014)*, Ankara, Turkey, 2014.

[38] C. L. Krumhansl, *Cognitive Foundations of Musical Pitch*. Oxford University Press, New York, 1990, ch. 4: A key-finding algorithm based on tonal hierarchies, pp. 77 – 110.

[39] K. K. Ganguli, "A corpus-based approach to computational modeling of melody in raga music," Ph.D. dissertation, Indian Institute of Technology Bombay, Mumbai, India, 2019.